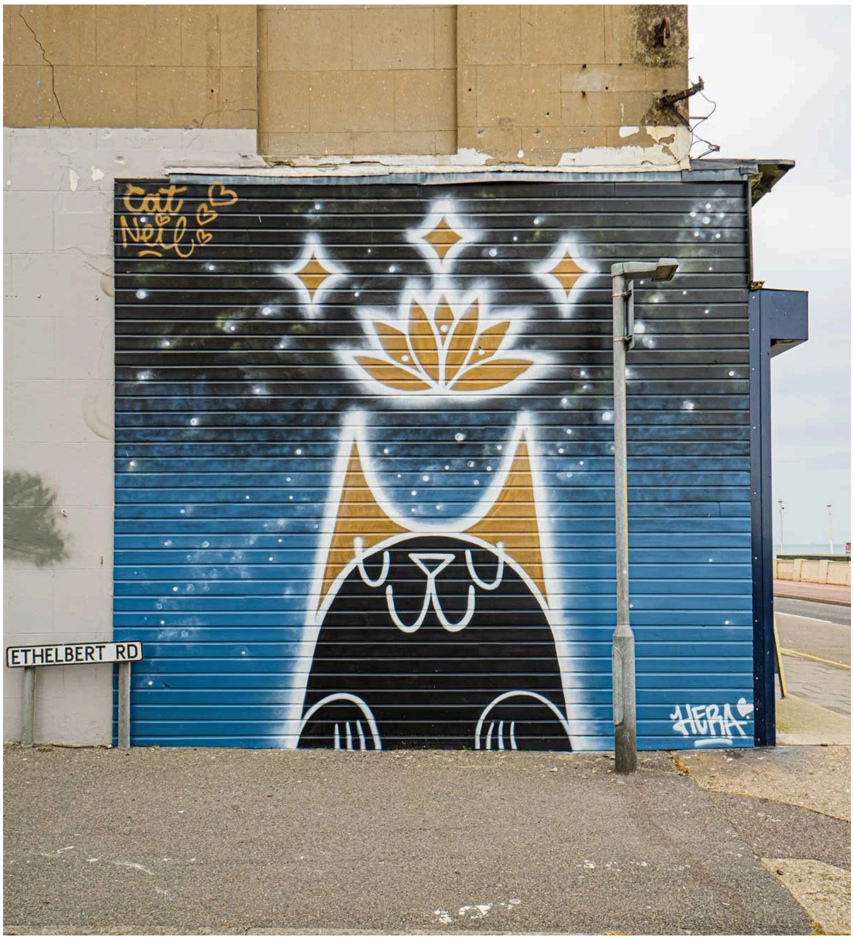

@ https://cenemagazine.co.uk/features/2021/6/4/catneil-kent-graffiti-artist

Graffiti in Vienna -

Reflection Paper

~1,975 words

ap224pu@student.lnu.se

(The DMP is attached below the reflection report, as it was created by DMP online.)

Introduction

The DMP for the “Graffiti Across Vienna” project draws inspiration from several sources, including, but not limited to, copyright laws, research ethics in academia, open science, and GDPR. The project is a tentative MA proposal, with its roots in a previous project that addressed graffiti in the 21st and 22nd districts of Vienna. The following reflection provides some thoughts on the DMP, which can be found after this paper: it illustrates the theoretical frameworks that guided its creation, as well as the challenges encountered in earlier projects of a similar nature.

Reflecting on the DMP is therefore not only a procedural requirement but also an opportunity to examine how theoretical principles, ethical considerations, and practical constraints shape the design of a DH project from the outset.

Ethics, FAIR Data, and Data Protection in DH

One of the more subtle and, thus, dangerous risks of Digitial Humanities, Katherine Hepworth and Christopher Church (2018) argue, is the assumption that digital projects are free from human prejudice and malice. Humans, after all, are not just the creators of algorithms, but also shape them. Consequently, digitisation projects should always factor in the potential for prejudices and biases to seep into the project, with the risk of unintentionally spreading misinformation or even causing harm (pp. 1-2). Nicolas Proferes (2020), in the same vein, states that research ethics stem from a multitude of different thinking schools, but can be summarised as treating people with respect, adhering to justice, and ensuring that anyone participating in the project benefits from it (p. 418).

In terms of *Graffiti Across Vienna*, the main focus will be placed on the graffiti, with the objective to display them as visual images on a 3D game, without any comment as to its artistic merit, the creator’s background or anything else that might take away from the main purpose of what it intends to do. More specifically, the metadata will focus on graffiti characteristics (e.g., location, capture data) rather than speculative elements such as the artist’s background, which risks perpetuating stereotypes. Bijan Rouhani (2023) believes that, while DH projects have tried to address ethical concerns, more work needs to be done in the area, as cultural heritage often necessitates answering questions of not only who is responsible for the project, how its metadata will be handled, but also how access to such data will shape visitors' perception of it (p. 3). This is particularly relevant for graffiti as a form of contested cultural heritage, where digital representation can influence how audiences interpret its social and political context. Additionally, these choices also highlight that metadata is not neutral; decisions about what to record and what to omit inevitably shape how the graffiti is interpreted and reused.

Graffiti Across Vienna occupies public space and, as a result, raises ethical concerns under data protection law. Personal data, which in public settings may include accidental captures of individuals’ faces or license plates, necessitates careful handling under GDPR, as the right to privacy is an undeniable human right (cf. European Commission, 2021, p. 4). The AoIR Ethical Guidelines similarly emphasise that publicly accessible data is not automatically free from ethical constraints, noting that visibility does not eliminate individuals’ expectations of contextual privacy, with young people in particular often assuming such protection (Franzke et al., 2020, p. 7). Indeed, the European Commission (2021) states that personal data must be anonymised to be GDPR-compliant (p. 6). In the interest of compliance, this project will either blur, crop, or entirely remove images that pose a risk of revealing personal data.

Any research project — be it scientific, humanities-based, or in between — benefits immensely from applying the FAIR principles, which help optimise data handling and, thus, enable the sharing of this data on an international and interdisciplinary scale (Betancort Cabrera et al., 2021, p. 2). These principles include findability, which ensures that not only humans, but also machines can read it (Betancort Cabrera et al., 2021, p. 4). G*raffiti Across Vienna* will enable this through metadata that follows the Dublin Core standard.

Tóth-Czifra (2020) notes that applying FAIR principles in the humanities can sometimes fail to yield genuinely FAIR outcomes due to legal issues around attribution, which create problems and make those responsible hesitant to release the data publicly (pp. 240-242). The second FAIR principle highlights the importance of accessibility: even if the data itself is lost, the metadata will remain, allowing scholars to understand what the project was about (Open Science EU, n.d.). As mentioned before, G*raffiti Across Vienna* uses standardised metadata but also stores metadata in Omeka, which runs in cloud-based environments.

Building on Prior Work: Lessons from Earlier Graffiti Documentation

The predecessor to this project is an Omeka-hosted collection documenting graffiti in the 21st and 22nd districts of Vienna. Like *Graffiti Across Vienna*, it used a smartphone to capture images and relied on Dublin Core for metadata, providing a useful foundation while also revealing several methodological limitations. These earlier experiences offer valuable insights that inform the current project.

In the predecessor, despite the use of Dublin Core, the metadata was minimal, exposing flaws. Figure 1 below highlights how the metadata, while present, remained basic and offered only limited contextual detail. Titles tended to be generic, descriptions were brief, and fields such as subject terms, location, dates, and rights information were recorded minimally. This reduced the depth and navigability of the collection, making it harder to interpret or reuse the images meaningfully.

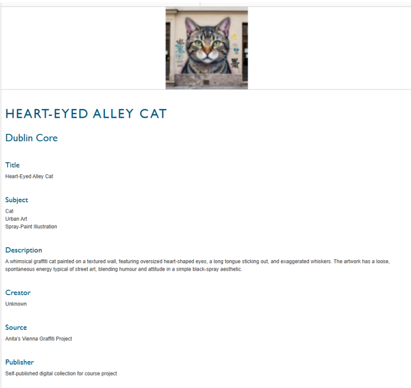

Figure 1

Screenshot of Omeka page.

Note. Screenshot.

As Anne J. Gilliland (2016, p. 1) notes, metadata is widely used yet frequently misunderstood, particularly as digitisation increases the need for consistent and accessible documentation. Similarly, Tóth-Czifra (2020, pp. 235–240) observes that humanities datasets often lose "thick description" when metadata is too sparse, resulting in flattened or decontextualised representations. The predecessor's minimal metadata likely reflected time constraints and the project's exploratory nature rather than a lack of methodological awareness. *Graffiti Across Vienna* will address this challenge by applying a standardised metadata schema that improves clarity, consistency, and overall usability.

Using the game *Oregon Trail* (1971) as an example, Trevor Owens (2018) argues that digitised items only remain relevant when they become persistent—when they accumulate a kind of cultural afterlife that, much like folklore, continues to engage new audiences long after their original creation. Owens also points to *Sid Meier's Civilisation IV: Colonisation*, where the game's code was publicly available, enabling users to examine how it was built, effectively turning it into part of a documented digital memory. Both cases illustrate his broader point: digital objects endure not simply because they exist, but because they are accompanied by enough contextual information to remain interpretable over time (pp. 45-48).

Graffiti Across Vienna will handle these challenges by implementing a richer, more consistent metadata schema that preserves contextual information and supports long-term interpretability. By documenting location, capture details, rights information, and descriptive elements in a structured and transparent way, the project ensures that the images remain meaningful not only in the present but also for future researchers who may revisit the dataset with new questions or methods.

Challenges, Trade-offs, and Methodological Decisions

The *Dublin Core Metadata Initiative* describes Dublin Core as a simple, general-purpose schema designed for broad interoperability rather than detailed cultural heritage description (DCMI, n.d.; cf. Gilliland, 2016, pp. 3–5). The simplicity of Dublin Core makes it easy to implement in small-scale DH projects, and it also aligns well with Omeka's built-in structure (cf. Gilliland, 2016). Furthermore, it supports FAIR principles through standardisation and machine-readability (cf. Betancort Cabrera et al., 2021, pp. 2–4). Yet, Dublin Core lacks the granularity needed to describe visual cultural heritage in depth, and cannot express technique, material, or stylistic relationships without custom fields (cf. Gilliland, 2016, pp. 6–8). Still, for a student-level, exploratory MA project, Dublin Core is proportionate and practical, and more expressive schemas (e.g., VRA Core, CIDOC-CRM) would exceed the project's scope (cf. Tóth-Czifra, 2020, pp. 240–242).

Documenting graffiti in public space risks capturing personal data such as faces or license plates, which must be handled carefully under GDPR (European Commission, 2021, pp. 4–6). At the same time, removing or altering images can reduce the dataset's contextual value, potentially affecting FAIR-aligned goals of reusability (cf. Tóth-Czifra, 2020, pp. 240–242). The AoIR guidelines emphasise that public visibility does not eliminate expectations of privacy (Franzke et al., 2020, p. 7), so this project prioritises anonymisation even when it limits the completeness of the visual record.

The project also faces practical constraints related to data capture and platform limitations. Smartphone photography can produce inconsistent lighting, angles, or image quality, particularly in outdoor environments, which may require additional post-processing to ensure the images are usable. This can extend the project's overall duration. Unity also requires balancing image resolution with performance to avoid excessive loading times, which affects how large or detailed the images can be. Solutions to improve performance fall outside the project's scope.

Limitations imposed by Unity also require selecting a consistent image export size before adding the images to Unity. Establishing this early avoids repeated re-processing and ensures that the files remain manageable throughout development. This helps maintain a predictable workflow without introducing unnecessary technical complications.

Conclusion

Overall, developing the DMP for *Graffiti Across Vienna* clarified how ethical, technical, and methodological considerations intersect in DH research. The project's focus on public-space graffiti initially suggested a straightforward documentation task, yet the preliminary work revealed multiple layers of responsibility, particularly regarding privacy, representation, and long-term data stewardship. Earlier experiences with the predecessor project highlighted the importance of consistent metadata and informed several improvements in the current approach. The DMP also underscores the value of transparency in research design, as documenting each step has made the project's limitations and assumptions more visible.

While the project operates within clear constraints—such as smartphone image quality, Unity performance limits, and the scope of a student-level study—it still aims to produce data that is usable, interpretable, and ethically sound. The reflection process has therefore reinforced the importance of careful planning and critical awareness in DH, demonstrating that even small projects benefit from structured data management and a clear articulation of their methodological choices.

Lastly, the process underscored how transferable these considerations are, as similar ethical and methodological questions will likely arise in future DH projects involving visual documentation or cultural heritage data.

AI Policy

The main utilisation of AI was restricted to the use of Grammarly.

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Plan Overview

A Data Management Plan created using DMPonline

Title: Graffiti Across Vienna

Creator: Anita Pal

Principal Investigator: Anita Pal

Data Manager: Anita Pal

Affiliation: Linnaeus University

Funder: Swedish Research Council

Template: Swedish Research Council Template

Project abstract:

Graffiti Across Vienna is a Unity 3D game that lets users experience a showcase of graffiti from all districts of Vienna by sitting in a tube and viewing them from the window. Using digitised images of graffiti captured by yours truly, the project documents and presents the city's diverse street art in a playful, immersive format. By bringing these photographs into a navigable 3D environment, the project aims to let users explore Vienna's graffiti culture remotely while also preserving a visual record of artworks that are often temporary.

ID: 204340

Start date: 15-01-2027

End date: 15-06-2027

Last modified: 15-05-2026

Graffiti Across Vienna

General Information

Graffiti Across Vienna

Anita Pal
ap224pu@student.lnu.se

204340

Version 1

15-05-2026

Description of data – reuse of existing data and/or production of new data

All images will be collected by the project creator using a smartphone and temporarily stored in Google Cloud for secure, redundant processing. Once processed, raw files will be deleted to comply with GDPR's data-minimisation principle. The processed images will then be uploaded to Omeka as the project's long-term dataset and reused in the Unity environment as the final, anonymised versions. No external datasets will be reused; all data is original to the project.

The project will produce digital photographs in JPEG or PNG format, which form the core dataset. A small amount of metadata will be created in Omeka using Dublin Core fields (e.g., Title, Date, Coverage, Description, Rights). All metadata will be stored in standard text-based formats. The total dataset is expected to be approximately 200–500 MB, depending on image resolution. No external data sources will be used.

Documentation and data quality

The material will be documented with clear file names and titles for each image, along with basic contextual metadata such as district, capture date, and a brief description. All metadata in Omeka will follow the Dublin Core standard, using fields such as Title, Date, Coverage, Description, and Rights. The collection method (smartphone photography in public space) will also be noted. Images reused in Unity will rely on the same Omeka metadata, and Unity assets will be kept in a simple, consistent folder structure without a separate metadata schema.

Data quality will be ensured through a consistent workflow. All images will be captured with the same smartphone and checked immediately for clarity, correct framing, and the absence of identifiable individuals. Filenames and metadata will be reviewed during processing to ensure accuracy and consistency. Any images that are blurry, incorrectly documented, or not fully anonymisable will be deleted. Because the dataset is small, quality control will be maintained through direct manual inspection.

Storage and backup

During the research process, all images will be stored in Google Cloud, which provides secure, redundant storage and protects the data if the smartphone is lost or damaged. Processed images and their metadata will also be stored in Omeka, which maintains server-level backups and preserves the metadata structure. Unity project files and processed images will be backed up through GitHub, adding version control and an additional layer of protection. Access to all storage locations will be restricted to the project creator.

Data security will be safeguarded by avoiding the collection of personal or sensitive data whenever possible. The project will follow GDPR-aligned practices by avoiding the capture of identifiable individuals, removing sensitive details when necessary, and deleting non-anonymisable images. All images and metadata will be stored in secure, access-controlled environments (Google Cloud, Omeka, and GitHub), with access restricted to the project creator through password-protected accounts and, where available, two-factor authentication. No data will be shared with third parties, and no raw files containing identifiable information will be retained, ensuring that both the data and its metadata remain protected throughout the project.

Legal and ethical aspects

Data handling will follow legal requirements by ensuring that no personal or sensitive data is collected or retained. Identifiable individuals will be avoided during capture, sensitive details will be blurred or cropped when necessary, and any images that cannot be anonymised will be deleted in line with GDPR's data-minimisation principle. Confidentiality will be maintained by storing all data in secure, access-controlled environments, with access restricted to the project creator and no sharing with third parties. Intellectual property rights will be safeguarded by using only images created by the project creator and by relying on standard, non-copyrighted Dublin Core metadata fields. Unity outputs will reuse only the project's own material and will not incorporate third-party assets.

Ethical data handling will be ensured by minimising risk and avoiding the collection of identifiable or sensitive information. Images will be captured in public spaces with care to avoid photographing people, and any unintentionally captured individuals or sensitive details will be blurred, cropped, or the image will be deleted. This follows ethical principles of respect, privacy, and minimisation. Access to the data will be limited to the project creator, and all material will be used solely for the project to avoid misrepresentation or unnecessary exposure. The project will rely entirely on original material created by the researcher, preventing ethical issues related to third-party ownership or misuse.

Accessibility and long-term storage

Processed images and their Dublin Core metadata will be publicly accessible on the project's Omeka site once documentation is complete, with no embargo. Only anonymised images will be shared; raw files are deleted after processing to comply with GDPR minimisation requirements. Core metadata fields (Title, Date, Coverage, Description) will be openly available.   Unity files may also be released via GitHub if they contain only original or openly licensed assets; files with restricted assets will remain private.   Reuse is permitted for non-commercial, academic, or educational purposes with attribution. Re-identification or reconstruction of deleted raw data is prohibited. No further restrictions apply.

Long-term storage will be managed by the project creator. Processed images and their Dublin Core metadata will be preserved in Omeka, which provides stable, backed-up storage suitable for long-term access. Only complete, anonymised, and relevant data will be retained; raw files are deleted after processing and will not be stored. Unity files may also be archived if they contain only original or openly licensed material, and if included, will be stored in a public GitHub repository or another suitable platform.

No specialised systems will be required for long-term access. The dataset consists of processed JPEG/PNG images and Dublin Core metadata, all of which are readable with standard tools. Omeka ensures long-term accessibility through browser-based access and exportable text-based metadata. If Unity files are shared, they will be optional supplementary materials; they can be opened with the free Unity Editor, but are not needed to understand or reuse the dataset. Overall, the data will remain platform-independent and usable without proprietary software.

The project does not require DOIs or other persistent identifiers. As a master 's-level project with a small dataset, long-term access will be provided through Omeka's stable item URLs and structured metadata, which are sufficient for citation and reuse. If Unity files are released, they may be hosted on GitHub, which provides stable versioned links. A formal DOI assignment is unnecessary for the project's scale, but could be added later if the dataset is deposited in a repository that supports DOI minting.

Responsibility and resources

All data management responsibilities, during the project and after its completion, will rest with the project creator. This includes organising, documenting, storing, backing up, and quality-controlling all data and metadata. After the project ends, the creator will continue to manage long-term storage by maintaining the Omeka collection and any public Unity or GitHub materials, ensuring they remain ethically, legally, and technically compliant over time.

Data management will require minimal resources. All storage, backup, documentation, and long-term preservation tasks will be handled by the project creator within the normal research workflow. Omeka will provide free long-term storage and public access for processed images and metadata; GitHub will offer free version control and backup for Unity files; Unity's free version is sufficient for integrating images into the game environment.   FAIR principles will be met through Omeka's stable URLs and structured Dublin Core metadata (Findability), browser-based access and common file formats (Accessibility), standard JPEG/PNG and text-based metadata (Interoperability), and clear documentation with rights-free images (Reusability). No additional costs, software, or infrastructure are required.